

4	(a)	(i)	Explain what is meant by the <i>diffraction</i> of a wave,	
				
				[1]
			Diffraction is the <u>spreading</u> of waves after passing through a slit or around the edge of an obstacle. B1 Comments: Many students did not remember this definition.	
		(ii)	state an important condition for significant diffraction to occur.	
				[1]
			For an observable pattern, the size of the obstacle should be of the same order in size as the wavelength of the light. B1 Comments: Many students did not remember this condition.	
	(b)		A diffraction grating with 300 lines per millimeter is being used in a typical light experiment. Different types of light are allowed to fall normally on a diffraction grating and the resultant pattern formed is to be studied. The first light source to be studied is a white light consisting of wavelengths between 400 nm and 700 nm	
		(i)	Find the maximum order of the complete spectrum that can be observed.	
				
			maximum order =	[2]
			$(1 \times 10^{-3}/300) \sin(90) > n(700 \times 10^{-9})$ $n < 4.76$ C1	

[Turn over

			Maximum order for each side of the grating is 4. A1 Comments: Many students did not remember that the 700nm wave is the limiting condition for the complete spectrum and use 400nm. It should be noted that bigger wavelength bends more.	
		(ii)	Find the order of the complete spectrum before the first overlapping between two higher order spectra.	
			order =	[3]
			For 1st order red $\theta = \sin^{-1} (700 \times 10^{-9}) / (3.333 \times 10^{-6}) = 0.2118 \text{ rad}$ For 2nd order red $\theta = \sin^{-1} 2(700 \times 10^{-9}) / (3.333 \times 10^{-6}) = 0.4339 \text{ rad}$ for 2nd order violet $\theta = \sin^{-1} 2(400 \times 10^{-9}) / (3.333 \times 10^{-6}) = 0.2426 \text{ rad}$ for 3rd order violet $\theta = \sin^{-1} 3(400 \times 10^{-9}) / (3.333 \times 10^{-6}) = 0.3686 \text{ rad} \quad \text{C1}$ Hence the 3rd order violet is overlapping in the 2nd spectrum. M1 There is only one pure spectrum. A1 Comments: Not well answered.	
	(c)		The next experiment is of light from a low pressure sodium lamp. Light from the lamp consists mostly of two wavelengths, 588.99 nm and 589.59 nm.	
		(i)	Explain quantitatively the problem that would likely arise in observing the spectral lines?	
				

				[2]
			Angle for 588.99nm, $\theta = \sin^{-1} (588.99 \times 10^{-9}) / (3.333 \times 10^{-6}) = 0.177809$ Angle for 589.59nm, $\theta = \sin^{-1} (589.59 \times 10^{-9}) / (3.333 \times 10^{-6}) = 0.177992$ C1 since the angle, between the two is very small, we might not be able to differentiate between the two lines. A1 Comments: Many students did not give a quantitative explanation despite being asked to do so.	
		(ii)	Suggest a refinement to the set up to help overcome this problem.	
				
				[1]
			To use a higher order for viewing, To change the grating to a finer one. B1 Comments: Not well answered.	
			[Total: 10]	

5	(a)	Define <i>magnetic flux density</i>	
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